Contrastive Representation Learning for Prompt–Response Alignment

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1 Introduction

LLMs are typically aligned with human preferences through reward models trained on pairwise comparisons. While effective, these approaches are known to suffer from systematic biases, including sensitivity to response length, formatting, and other superficial features.

A central limitation of current alignment methods is the absence of an explicit representation of *prompt–response alignment*. Instead, alignment is implicitly encoded in model parameters, making it difficult to interpret and control.

This proposal aims to address this gap by building a shared representation space for prompt and response embeddings via contrastive learning, learning the features of chosen responses of corresponding prompts compared with rejected ones, analyzing the geometric structure of the shared shared representation learning space between prompt and response embeddings, and developing a robust retrieving method for downstream RAG tasks.

Research Question. Can contrastive representation learning be used to learn an explicit, robust, and transferable representation of prompt–response alignment that captures user intent while remaining invariant to spurious artifacts? Can we analyze the geometric structure of the prompt-response shared representation space? Is there any linear or linear relationship between aligned and unaligned representations? Can we find out how DPO changes the representation space structure?

2 Related Work

2.1 Reward Modeling and RLHF

Reinforcement Learning from Human Feedback (RLHF) has become a widely adopted framework for aligning large language models (LLMs) with human preferences [Christiano et al., 2017]. However, recent work has revealed that reward models are susceptible to reward hacking, where models exploit imperfections in the learned reward function rather than genuinely aligning with human intent [Amodei et al., 2016]. Reward hacking can arise from unintentional and prompt-irrelevant human preferences [Wang et al., 2025].

2.2 Contrastive Representation Learning

Contrastive Language–Image Pre-training (CLIP) [Radford et al., 2021] learns aligned representations across modalities via a contrastive objective. Using a dual-encoder architecture, CLIP maps images and texts into a shared embedding space, where matched pairs are encouraged to have high similarity and mismatched pairs are pushed apart. This approach provides a general framework for learning semantic alignment through contrastive representation learning.

2.3 Representation Learning for Alignment

Contrastive representation learning has been theoretically characterized by two key principles: *alignment*, which encourages representations of positive pairs to be close, and *uniformity*, which promotes a well-distributed representation space [Wang and Isola, 2020]. This perspective provides a useful foundation for our setting, where prompt–response alignment can be viewed as learning representations that preserve semantic correspondence while remaining robust to spurious correlations.

3 Intuition

We do not view alignment as a matter of removing all surface-level features. The harder problem is to determine when a feature is genuinely related to the prompt and when it is just noise. In preference data, the same feature can be helpful in one context and misleading in another. For example, we always treat length as a spurious artifact and aim to remove length bias. However, when a prompt mentioned "giving more details", it includes an intention of getting longer responses.

Our contrastive framework asks a different question: for a given prompt, which responses actually reflect its intention? For each prompt p_i, we collect four responses from different LLMs and treat each matched pair (p_i, r_{ij}) as positive, while treating mismatched pairs $(p_i, r_{kj}), k \neq i$, as negative. This shifts the learning problem away from fitting broad preference patterns and toward modeling prompt-specific compatibility. Because the negatives come from other prompts and the positives come from diverse models, superficial features that are not tied to the prompt are less likely to serve as reliable shortcuts. What remains more consistent across positive pairs is whether the response captures the intention of the prompt. The resulting similarity is therefore meant to reflect not absolute response quality, but how much a response expresses the meaning and intent of its prompt.

4 Proposed Framework

4.1 Model Architecture and Training Strategy

We adopt a dual-encoder architecture consisting of a prompt encoder and a response encoder. Each encoder is composed of two components: a pre-trained language model and a learnable transformation module.

Specifically, for a prompt p and a response r, we first obtain initial representations using a frozen

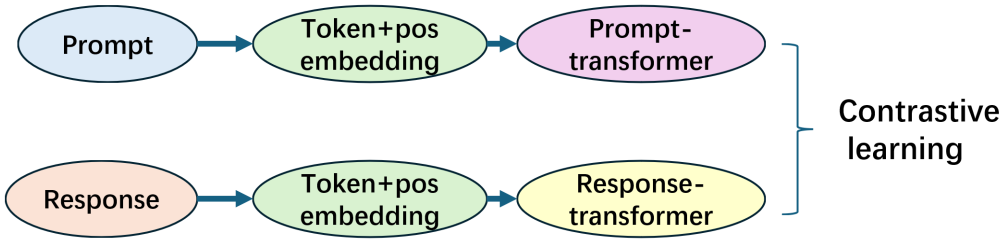

Figure 1: Contrastive Learning Framework

GPT-2 encoder:

$$h_p = \text{GPT2}(p), \quad h_r = \text{GPT2}(r).$$

These representations are then passed through two independent Transformer-based projection networks:

$$f(p) = T_p(h_p), \quad g(y) = T_r(h_r),$$

where T_p and T_r denote the prompt and response Transformer modules, respectively. The two Transformer modules do not share parameters.

During training, we freeze the parameters of the GPT-2 encoders and only update the Transformer projection networks. This design isolates the learning process to the alignment layer, encouraging the model to learn task-specific alignment representations without altering the underlying language representations.

4.2 Contrastive Objective

For each prompt p_i, we collect four responses generated by different large language models, denoted by

$$\{r_{i1}, r_{i2}, r_{i3}, r_{i4}\}.$$

This gives a total of $4N$ matched prompt–response pairs for N prompts. Unlike standard preference learning, our objective is not to directly model which response is preferred over another, but to learn a similarity measure that reflects how much prompt-specific intention is expressed in a response.

We define the similarity between a prompt p_i and a response r_{kj} as

$$s_{i,kj} = \frac{f(p_i)^\top g(r_{kj})}{\tau},$$

where $f(\cdot)$ and $g(\cdot)$ are the prompt and response encoders, respectively, and τ is a temperature parameter.

For each prompt p_i, all matched responses $\{r_{i1}, r_{i2}, r_{i3}, r_{i4}\}$ are treated as positive examples. Responses associated with other prompts,

$$\{r_{kj} : k \neq i,\ j \in \{1, 2, 3, 4\}\},$$

are treated as negative examples. Under this construction, the primary systematic difference between

positive and negative pairs is whether the response contains information relevant to the intention of the given prompt. Since the responses are generated by multiple LLMs and the negatives are drawn from other prompts, superficial artifacts such as length, formatting, or model-specific style are less likely to provide a stable shortcut across all training pairs.

We optimize an InfoNCE-style contrastive loss. For each positive pair (p_i, r_{ij}), the model is trained to assign higher similarity to the matched response than to responses associated with other prompts:

$$\mathcal{L} = -\frac{1}{4N} \sum_{i=1}^{N} \sum_{j=1}^{4} \log \frac{\exp(s_{i,ij})}{\sum_{k=1}^{N} \sum_{m=1}^{4} \exp(s_{i,km})}.$$

Equivalently, for each anchor prompt p_i, the numerator corresponds to one of its matched responses r_{ij}, while the denominator includes all candidate responses in the batch.

This objective encourages the model to bring a prompt closer to responses that encode its semantic content and intention, while pushing it away from responses generated for other prompts. After training, responses that lie closer to p_i in the shared embedding space can be interpreted as containing more information about the intention behind p_i.

5 Experimental Plan

5.1 Datasets and Training Setup

We will construct training examples from preference datasets containing multiple candidate responses per prompt. For each prompt p_i, we collect four responses generated by different large language models, denoted by $\{r_{i1}, r_{i2}, r_{i3}, r_{i4}\}$. Each matched pair (p_i, r_{ij}) is treated as a positive example. For a given prompt p_i, responses associated with other prompts, (p_i, r_{kj}) for $k \neq i$, are treated as negative examples. This design yields $4N$ positive prompt–response pairs for N prompts, together with in-batch negatives drawn from mismatched prompts.

We will use the RLHF preference dataset from RLHFlow as the primary data source and split the data into training, validation, and test sets by prompt so that prompts in the test set are unseen during training. This prevents leakage of prompt-specific information across splits and ensures that evaluation measures generalization to new prompts rather than memorization of training pairs.

5.2 Baselines

We will compare the proposed contrastive framework against the following baselines:

1. **Frozen embedding similarity:** cosine similarity between prompt and response embeddings obtained directly from the frozen pretrained encoder, without contrastive training.
2. **Standard reward model:** a preference model trained to assign higher scalar scores to chosen responses than rejected responses.

5.3 Experiment 1: Prompt–Response Retrieval Accuracy

Our first experiment evaluates whether the learned embedding space captures prompt-specific semantic compatibility. Given a prompt p_i, we rank a candidate pool consisting of its four matched responses and a set of mismatched responses sampled from other prompts. We then measure whether the responses associated with p_i are assigned higher similarity than mismatched responses.

We will report the following retrieval metrics:

1. **Recall@K:** whether at least one matched response appears in the top-K retrieved candidates.
2. **Mean Reciprocal Rank (MRR):** the reciprocal rank of the highest-ranked matched response.
3. **Average matched-pair similarity gap:**

$$\Delta_{\text{match}} = \frac{1}{4N} \sum_{i=1}^{N} \sum_{j=1}^{4} \left[s(p_i, r_{ij}) - \frac{1}{|\mathcal{N}_i|} \sum_{r \in \mathcal{N}_i} s(p_i, r) \right],$$

   where $\mathcal{N}_i$ denotes the set of sampled mismatched responses for prompt p_i.

This experiment tests the core objective of the method: whether the contrastive model learns to identify responses that contain information specific to the corresponding prompt.

5.4 Experiment 2: Controlled Length-Bias Evaluation

To evaluate sensitivity to response length, we will design a controlled experiment in which prompt intention is held approximately fixed while response length varies. For a subset of prompts, we will construct or collect pairs of responses that are similar in semantic content but differ substantially in length, such as a concise answer and an expanded version of the same answer.

For each prompt p, let r_{short} and r_{long} denote two responses with similar semantic content but different lengths. We will compare the similarity assigned by the contrastive model and the scores assigned by reward model baselines. We will report:

1. **Length-sensitivity gap:**

$$\Delta_{\text{len}} = s(p, r_{\text{long}}) - s(p, r_{\text{short}}).$$

2. **Correlation with response length:** the Pearson and Spearman correlation between model score and response length across candidate responses for the same prompt.

If the learned similarity is less driven by prompt-independent verbosity, the contrastive model should exhibit a smaller length-sensitivity gap and a lower score–length correlation than standard reward models.

5.5 Experiment 3: Counterfactual Surface-Feature Perturbation

A central claim of this project is that the learned similarity should reflect prompt intention rather than superficial response artifacts. To test this directly, we will create counterfactual response variants that preserve the main semantic content of a response while altering surface-level properties. These perturbations may include paraphrasing, formatting changes, lexical substitutions, and controlled stylistic expansion or compression.

Given an original response r and a perturbed version r' for the same prompt p, we will measure the stability of the learned representation under such perturbations. Specifically, we will report:

1. **Similarity stability:**
$$\Delta_{\text{cf}} = |s(p, r) - s(p, r')|.$$

2. **Embedding displacement:**
$$d_{\text{cf}} = \|g(r) - g(r')\|_2.$$

These quantities will be compared across different types of perturbations. Small changes under surface-level perturbations would suggest that the model is capturing prompt-conditioned semantic compatibility rather than overfitting to lexical or stylistic details.

5.6 Experiment 5: Representation Analysis

To understand what information is encoded in the learned embedding space, we will conduct a series of post hoc analyses.

Geometric analysis. We will apply PCA and related dimensionality-reduction methods to visualize the shared representation space. In particular, we will examine whether dominant embedding directions are strongly associated with response length, lexical overlap, or generator identity. If such directions exist, we will quantify their relationship with model scores and assess whether alignment-relevant structure remains after controlling for them.

5.7 Evaluation Criteria

Taken together, these experiments evaluate the proposed framework along three dimensions:

1. **Alignment fidelity:** whether the learned similarity retrieves prompt-matched responses and separates them from mismatched ones;

2. **Robustness to spurious artifacts:** whether the model is less sensitive to length and other surface-level perturbations than standard reward models;

3. **Representation quality:** whether the embedding space exhibits prompt-conditioned structure that is not dominated by superficial correlates.

A successful outcome would demonstrate that contrastive representation learning provides a useful representation-level perspective on alignment, and that the resulting similarity captures the extent to which a response reflects prompt-specific intention.

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